

Dimensional Analysis and Significant Figures

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Part I

Dimensional Analysis

Problem Set

Chapter 1

JEE Mains Problems

Part II

Significant Figures

Problem Set

Chapter 2

JEE Mains Problems

Q1 : The diameter of a wire measured by a screw gauge of least count 0.001 cm is 0.08 cm. The length measured by a scale of least count 0.1 cm is 150 cm. When a weight of 100 N is applied to the wire, the extension in length is 0.5 cm, measured by a micrometer of least count 0.001 cm. The error in the measurement of Young's Modulus is $\alpha \times 10^9 \text{ N/m}^2$. The value of α is

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(Ignore the contribution of load to Young's Modulus error calculation)

[JEE Mains 2nd April 2026 Shift 1]

- A) 1.3
- B) 1.65
- C) 0.13
- D) 0.25